

# Soma Native

Author: Dalin Jia (362118251@qq.com)

Affiliation: Independent Researcher

Email: 362118251@qq.com

Date: June 2026

Version: v3.0

## Soma Native

### Soma Native Architecture: A Signal Field-Native Neural Network Design

#### Abstract

#### 1. Introduction

##### 1.1 Transformer

##### 1.2

##### 1.3 Soma Native

#### 2. Method

##### 2.1

$$S(x, t) = \sum_{\tau < t} \gamma^{t-\tau} \cdot a(x, \tau)$$

$$\text{SF}(q, K, V) = \text{Attention}_{\text{near}}(q, K_{\text{ring}}, V_{\text{ring}}) + \alpha \cdot \text{Attention}_{\text{far}}(q, S_K, S_V)$$

2.2 Unified Field Block

$$\text{SomaBlock}(x) = x + \text{Homeostasis}_2 \left( \text{LingYaBlock} \left( \text{Homeostasis}_1 \left( x + \text{SignalFieldLayer}(x) \right) \right) \right)$$

2.3 Homeostasis

$$\text{Homeostasis}(x)_i = x_i \cdot \rho_i$$

3. Experiments

3.1

3.2

3.3

| Transformer Attention | Soma SignalField |        |         |
|-----------------------|------------------|--------|---------|
| 128                   | ~65K             | ~49K   | **24%** |
| 256                   | ~262K            | ~197K  | **25%** |
| 512                   | ~1.05M           | ~786K  | **25%** |
| 768                   | ~2.36M           | ~1.77M | **25%** |

3.4 FLOPs

\*\* 7SFA vs Standard Attention FLOPsseq=1024, d=512\*\*

| SFA   | Standard Attention   |                      |             |
|-------|----------------------|----------------------|-------------|
| FLOPs | 1.08×10 <sup>9</sup> | 1.61×10 <sup>9</sup> | ** -32.8%** |
| 注意力   | 8.9×10 <sup>6</sup>  | 5.4×10 <sup>8</sup>  | ** -98.4%** |

3.5

| Standard Attention |                 |              |                                    |
|--------------------|-----------------|--------------|------------------------------------|
| 64                 | 12.5 $\mu$ s    | 11.8 $\mu$ s | -5.6%                              |
| 1,024              | 45.0 $\mu$ s    | 11.8 $\mu$ s | <b>**3.8<math>\times</math>**</b>  |
| 8,192              | 360.0 $\mu$ s   | 11.8 $\mu$ s | <b>**30.5<math>\times</math>**</b> |
| 65,536             | 2,880.0 $\mu$ s | 11.8 $\mu$ s | <b>**244<math>\times</math>**</b>  |

### 3.6 7

| Soma Native | Transformer        |        |                                   |
|-------------|--------------------|--------|-----------------------------------|
|             | 462 KB $\times$ 28 | 114 MB | <b>**248<math>\times</math>**</b> |
|             | TBD                | TBD    | —                                 |
| 验证集         | <b>**TBD**</b>     | 6.66   | —                                 |

## 4. Discussion

### 4.1 Mamba

| Mamba SSM | Soma Native    |                        |
|-----------|----------------|------------------------|
|           |                | +                      |
|           | $O(n \cdot d)$ | $O(k \cdot n \cdot d)$ |
|           | $O(d)$         | $O(k \cdot d)$         |
|           | ✓              | ✓                      |
|           | $\times$       |                        |

### 4.2

### 4.3

## 5. Conclusion

## References

[1] Vaswani A, et al. Attention Is All You Need. \*NeurIPS\*, 2017.

[2] Gu A, Dao T. Mamba: Linear-Time Sequence Modeling with Selective State Spaces. \*arXiv:2312.00752\*, 2023.

[3] Katharopoulos A, et al. Transformers are RNNs: Fast Autoregressive Transformers with Linear Attention. \*ICML\*, 2020.

[4] Dao T, et al. FlashAttention: Fast and Memory-Efficient Exact Attention with IO-Awareness. \*NeurIPS\*, 2022.

[5] Chen S, et al. RetNet: Retentive Network: A Successor to Transformer for Large Language Models. \*arXiv:2307.08621\*, 2023.

[6] Zhang B, Sennrich E. Root Mean Square Layer Normalization. \*NeurIPS\*, 2019.

[7] Su J, et al. RoFormer: Enhanced Transformer with Rotary Position Embedding. \*arXiv:2104.09864\*, 2021.